Macroeconomics I — MPE Insper

Practice list: Consumption, Saving and Labor Markets — qualifier level (Listas 3–4)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Block 1 — Consumption, present bias and interest rates

(Kurlat chs. 6, 9; Romer ch. 8 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A household lives $T+1$ periods ($t = 0, \ldots, T$; $T \geq 1$) with $u(c) = c^{1-\sigma}/(1-\sigma)$, $\sigma > 0$ ($\ln c$ if $\sigma = 1$), discount factor $0 < \beta < 1$ and a constant real rate $r > 0$; its total wealth (financial plus human, in present value at $t = 0$) is W. The share c_0/W is:

A. $(1-\gamma)/(1-\gamma^{T+1})$ with $\gamma = \beta^{\sigma}(1+r)^{\sigma-1}$

B. $(1-\gamma)/(1-\gamma^{T+1})$ with $\gamma = \beta^{1/\sigma}(1+r)^{1/\sigma-1}$

C. $(1-\gamma)/(1-\gamma^{T+1})$ with $\gamma = \beta$

D. $(1-\gamma)/(1-\gamma^{T+1})$ with $\gamma = [\beta(1+r)]^{\sigma}$

b) (3 points) *Consumption × general equilibrium.* A two-period endowment economy with a representative household: endowments $y_1 > 0$ and $y_2 = (1+g)y_1$, $g > 0$; preferences $u(c_1) + \beta u(c_2)$, CRRA with $\sigma > 0$, $0 < \beta < 1$; a bond in zero net supply pays r. The equilibrium real interest rate is:

A. $r = (1+g)/\beta - 1$

B. $r = 1/\beta - 1$

C. $r = (1+g)^{1/\sigma} - 1$

D. $r = (1+g)^{\sigma}/\beta - 1$

c) (2 points, T or F) *Consumption × credit markets.* Two periods, endowments (y_1, y_2), utility $u(c_1) + \beta u(c_2)$ with u strictly increasing and strictly concave; the household saves at r_s and borrows at $r_b > r_s$. At the current rates it neither borrows nor lends, and the first unit saved would strictly lower its utility at the margin. Then any rise in r_s, however small, makes it save.

d) (2 points, T or F) *Consumption × fiscal policy.* Two periods, utility $u(c_1) + \beta u(c_2)$ with u strictly increasing and strictly concave, income y_t, lump-sum taxes τ_t; the household's saving faces the limit $a \geq -b$, $b \geq 0$, which binds. A tax cut $\Delta > 0$ today is financed by a tax increase $(1+r)\Delta$ next period. Then current consumption rises by exactly Δ, for every $\Delta > 0$.

Question 2. (15 points) Discursive. *Consumption × behavioral economics × general equilibrium.* Three periods $t = 0, 1, 2$; in (a)–(c) the real interest rate is zero and the household has wealth $W > 0$ at date 0. The self at date t maximizes $u(c_t) + \beta \sum_{s>t} u(c_s)$ with $u(c) = c^{1-\sigma}/(1-\sigma)$, $\sigma > 0$ ($\ln c$ if $\sigma = 1$), and $0 < \beta < 1$.

a) (3 points) Find the plan chosen with commitment (self 0 chooses all three consumptions), by a naive self 0 (who believes future selves will follow its plan) and by a sophisticated self 0 (who anticipates self 1's choice). In what sense are these preferences time-inconsistent?

Solution:

b) (3 points) Determine when the sophisticated self 0 consumes more, less or the same as the naive one, and explain the mechanism.

Solution:

c) (3 points) At date 0 self 0 can buy an illiquid account that locks in the (c_1, c_2) it chooses. What share x of W would a sophisticated self 0 pay for it at most? Show whether x depends on W, bound it, give it under log utility and find how it behaves as $\beta \to 1$. What would a naive self 0 pay?

Solution:

d) (3 points) Now each household receives $y > 0$ at each date and has no initial wealth; households are identical and sophisticated, and borrow and lend at r_{01} (from date 0 to 1) and r_{12} (from 1 to 2). Find the equilibrium interest rates, show whether they depend on σ, and draw them against β.

Solution:

e) (3 points) *Parameter analysis.* Sign the effects of β, of σ and of naivete (all households naive

instead of sophisticated, also in (d)) on x, r_{01} and r_{12}; prove each sign and collect the signs in a table, give the limits of x at the ends of the admissible ranges, and read the results.

Solution:

Block 2 — Labor supply, taxes, transfers and search

(Kurlat ch. 7; Romer ch. 10 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Labor × consumption smoothing.* A household lives $N \geq 2$ periods with per-period utility $\ln c_t - \psi\, h_t^{1+1/\varepsilon}/(1+1/\varepsilon)$, $\psi, \varepsilon > 0$; it has no initial wealth, $\beta = 1$ and $r = 0$, and the wage is the same in every period. The wage rises by the same small proportion in k of the periods, $1 \leq k < N$. To first order, the elasticity of hours in a period with the raise with respect to its wage is:

A. ε

B. $\varepsilon/(1+\varepsilon)$

C. $\varepsilon\,(N-k)/N$

D. $\varepsilon\,(N-k)/(N+k\varepsilon)$

b) (3 points) *Search × wage setting.* Steady-state search model with matches $M = A_m U^{\xi} V^{1-\xi}$, $\xi \in (0,1)$, tightness $\theta = V/U$, vacancy-filling rate $q(\theta) = M/V$ and free entry $(y-w)\,q(\theta) = (r+s)\kappa$, with $r, s, \kappa > 0$. The wage is rigid at $w \in (0, y)$ while productivity y moves. The elasticity of θ with respect to y is:

A. $\frac{1}{\xi}\,\frac{y}{y-w}$

B. $\frac{1}{1-\xi}\,\frac{y}{y-w}$

C. $\frac{1}{1-\xi}$

D. $\frac{1-\xi}{\xi}\,\frac{y}{y-w}$

c) (2 points, T or F) *Search × efficiency.* One-shot matching: U unemployed workers and V vacancies, each costing $\kappa > 0$ to open (free entry), form $M(U,V)$ matches, with M of constant returns, increasing and strictly concave in V. A match produces y, of which the worker gets the share $\phi \in (0,1)$. The equilibrium maximizes net output $yM(U,V) - \kappa V$ if and only if ϕ equals the elasticity of M with respect to U at the equilibrium tightness.

d) (2 points, T or F) *Labor × transfers.* With $u = \ln c + \alpha \ln(1-h)$, $\alpha > 0$, a household that receives a lump-sum transfer $T > 0$ (taken as given) works the same hours at every after-tax wage w_n.

Question 4. (15 points) Discursive. *Labor supply × taxes and transfers × welfare.* Identical households have $u = \ln c + \alpha \ln(1 - h)$, $\alpha > 0$, and one unit of time; a linear technology pays a wage of one per hour. Labor income is taxed at $\tau \in [0, 1)$ and the revenue is returned as a lump-sum transfer T that each household takes as given.

a) (3 points) Derive individual labor supply given (τ, T). Show what the tax does when $T = 0$, and find equilibrium hours $h(\tau)$ when the budget balances.

Solution:

b) (3 points) Find the tax rate τ^* that maximizes revenue $R(\tau)$ (the size of the transfer program) and the maximal revenue $R^* \equiv R(\tau^*)$, both in closed form; prove that revenue is single-peaked, and draw $R(\tau)$ for two values $\alpha < \alpha'$.

Solution:

c) (3 points) Derive in closed form the welfare cost of the program in consumption-equivalent terms: the share $x(\tau)$ of consumption at $\tau = 0$ the household would give up to avoid the program. Determine its order at $\tau = 0$ and its monotonicity in τ; find its leading term around $\tau = 0$ and its limit as $\tau \to 1$.

Solution:

d) (3 points) At the balanced-budget allocation, for $\tau \in (0,1)$, derive the Marshallian and Hicksian elasticities of individual hours with respect to the after-tax wage $w_n = 1-\tau$, the balanced-budget elasticity $e_{BB} \equiv d\ln h / d\ln(1-\tau)$, and the Frisch elasticity. Order the four for every $\tau \in (0,1)$ and say which one Prescott's experiment uses.

Solution:

e) (3 points) *Parameter analysis.* Sign the effects of α and of τ on h, R, τ^*, R^*, x and e_{BB}, prove each sign and collect them in a table. Find the limits as $\alpha \to 0$ and $\alpha \to \infty$ and a bound on τ^* valid for every α, and read the results.

Solution:

Block 3 — Integrative: consumption, labor, pensions and welfare

(Kurlat chs. 2, 4, 6–7; Romer chs. 2, 8, 10 • 25 points)

Question 5. (10 points) Objective items.

a) (3 points) *Consumption × fiscal policy × credit.* Brazil's 2020 emergency aid, stylized. Households live two periods with utility $u(c_1) + \beta u(c_2)$ (u strictly concave, $\beta \in (0,1)$) and save a at the real rate $r > -1$ at which the government borrows, subject to a fixed limit $a \geq -b$, $b \geq 0$. A share $\lambda \in (0,1)$ of households is at the limit; the other $1-\lambda$ are strictly inside it. Every household gets a lump-sum transfer $T > 0$ in period 1, financed by debt repaid with interest through equal lump-sum taxes in period 2; the limit still binds for the constrained after the transfer. At the given r, average period-1 consumption per household changes by:

A. $\Delta\bar{c}_1 = T$

B. $\Delta\bar{c}_1 = (1-\lambda)T$

C. $\Delta\bar{c}_1 = 0$

D. $\Delta\bar{c}_1 = \lambda T$

b) (3 points) *Labor × welfare measurement.* Jones and Klenow's flow utility (Kurlat §2.2, with $\sigma = 1$) is $\ln c - \theta(1-l)^2$, $\theta > 0$, where l is leisure and $1 - l = h$ is market hours. Between two dates, consumption per person in Brazil moves from c_0 to c_1 and market hours from h_0 to h_1 (both in $(0, 1)$), with survival and inequality unchanged. The consumption equivalent Λ, the factor on date-0 consumption (hours held at h_0) that delivers date-1 welfare, satisfies:

A. $\ln \Lambda = \ln(c_1/c_0)$

B. $\ln \Lambda = \ln(c_1/c_0) + \theta(h_0^2 - h_1^2)$

C. $\ln \Lambda = \ln(c_1/c_0) + \theta(h_0 - h_1)$

D. $\ln \Lambda = \ln(c_1/c_0) - \theta(h_0 - h_1)$

c) (2 points, T or F) *Consumption × labor.* A household maximizes $\sum_{t=0}^{N} \beta^t [\ln c_t - v(h_t)]$ ($N \geq 1$, v increasing and strictly convex, interior hours) and borrows and lends freely at a given rate r. With these separable preferences, news at $t = 0$ of a permanent wage increase from $t = 1$ on, with w_0 unchanged, makes the household work more at $t = 0$.

d) (2 points, T or F) *Search × social insurance.* In the steady-state search model (tightness θ, vacancy-filling rate $q(\theta)$ and job-finding rate $f(\theta) = \theta q(\theta)$ with $q' < 0 < f'$, productivity y, unemployment benefit $z < y$, flow vacancy cost κ, separation rate s, interest rate r, free entry of vacancies, Nash-bargained wage with workers' bargaining power $\phi \in [0, 1)$), more generous unemployment benefits (a higher z, e.g. a more generous *seguro-desemprego*) lower job creation θ even when workers have no bargaining power ($\phi = 0$).

Question 6. (15 points) Discursive. *Integrative: life-cycle saving × labor supply × pension design × the interest–growth gap.* Brazil's general regime (INSS) is pay-as-you-go (PAYG), with benefits computed from each worker's contribution record, while the minimum-wage floor and the non-contributory BPC pay amounts unrelated to it; the 2019 reform proposal included funded individual accounts (*capitalização*), dropped in Congress. A household lives two periods: in period 1 it works $h \in (0, 1)$ at the wage $w > 0$, in period 2 it is retired. Utility is $u = \ln c_1 + \gamma \ln(1 - h) + \beta \ln c_2$, with $\beta \in (0, 1)$ and $\gamma > 0$. A payroll tax $\tau \in (0, 1)$ finances a pension P; the budgets are $c_1 + a = (1 - \tau)wh$ and $c_2 = (1 + r)a + P$, with a free in sign. Each generation is $1 + n$ times the previous one, wages are constant, $n, r > -1$, prices (r, w) are taken as given in (a), (b) and (d), and $r > n$ unless stated; write $q \equiv \frac{1+n}{1+r}$. Three designs: *funded*, $P = (1 + r)\tau wh$; *Bismarckian PAYG*, $P = (1 + n)\tau wh$, linked to the household's own contributions (it internalizes the link); *Beveridgean PAYG*, a lump sum P that the household takes as given, equal to $(1 + n)\tau wh$ in equilibrium.

a) (3 points) *Funded and Bismarckian designs.* Write each lifetime budget as $c_1 + \frac{c_2}{1+r} = \omega wh$ and find ω. Solve the household problem (both consumptions, and hours h_F and h_{Bis}) and compare the two designs.

Solution:

b) (3 points) *Beveridgean design.* Find equilibrium hours h_{Bev}, compare them with h_F, explain the mechanism, and draw hours against $\tau \in (0,1)$ for the three designs.

Solution:

c) (3 points) *Saving and capital.* Find private saving per unit of wage income, $a/(wh)$, in both PAYG designs, and the national saving rates s_F, s_{Bis} and s_{Bev} (private saving plus pension assets, per unit of wage income); rank them. If these are the saving rates of a Cobb–Douglas Solow economy with capital share $\alpha \in (0,1)$, the same n, technology growth g and depreciation δ ($\delta + n + g > 0$), by what factor does each PAYG design change steady-state output per effective worker relative to the funded design?

Solution:

d) (3 points) *Welfare.* At given prices, measure each PAYG design by its consumption-equivalent loss λ relative to funded: the share of funded consumption in both periods, at funded hours, whose loss leaves the household as well off as under the PAYG design. Find λ_{Bis} and λ_{Bev} in closed form, rank the three designs when $r > n$ (the two PAYG designs by a revealed-preference argument), and determine what changes when $r = n$ and when $r < n$.

Solution:

e) (3 points) *Parameter analysis.* Sign the effects of r, n and τ on ω, h_{Bev}, s_{Bis}, s_{Bev}, the two Solow output ratios of (c), λ_{Bis} and λ_{Bev}; prove each sign and collect the signs in a table; find the limits $\tau \to 0$ (with the order in τ of $\lambda_{Bev} - \lambda_{Bis}$) and $r \to n$, and read the results.

Solution: